DSA0605 – Data Handling and Visualization

CO2-AT1- Chart Construction Test

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Q1. Boxplots for Distribution Comparisons

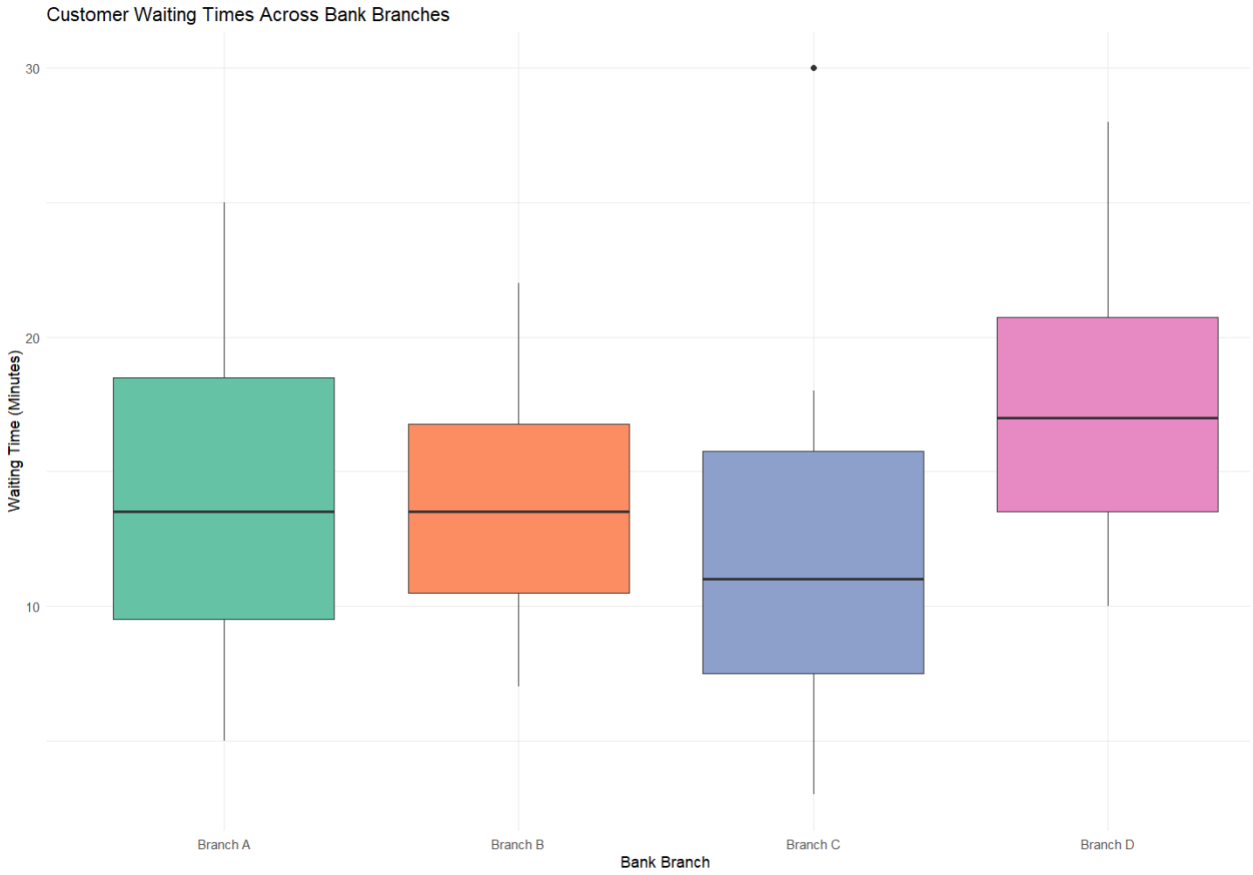
Aim: Compare customer waiting-time distributions across four bank branches.

```
library(ggplot2)
waiting <- data.frame(
  Branch = rep(c("Branch A", "Branch B",
                 "Branch C", "Branch D"), each = 8),
  WaitingTime = c(
    5, 8, 10, 12, 15, 18, 20, 25,
    7, 9, 11, 13, 14, 16, 19, 22,
    3, 6, 8, 10, 12, 15, 18, 30,
    10, 12, 14, 16, 18, 20, 23, 28
  )
)
ggplot(waiting, aes(x = Branch, y = WaitingTime,
                    fill = Branch)) +
  geom_boxplot() +
  labs(
    title = "Customer Waiting Times Across Bank Branches",
    x = "Bank Branch",
    y = "Waiting Time (Minutes)"
  ) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal() +
  theme(legend.position = "none")
```

Interpretation:

- The boxplots compare the median, spread and variability of waiting times.
- Branch C has the widest overall range in the sample.

- Branch D generally has higher waiting times than Branches A and B.
- Potential outliers can be identified from points beyond the whiskers.



Conclusion: Boxplots are appropriate for comparing distributions and identifying differences in waiting times among bank branches.

Q2. Amounts – Total Sales Revenue

Aim: Display the total annual sales revenue of five sales representatives using a bar chart.

```
library(ggplot2)

sales <- data.frame(

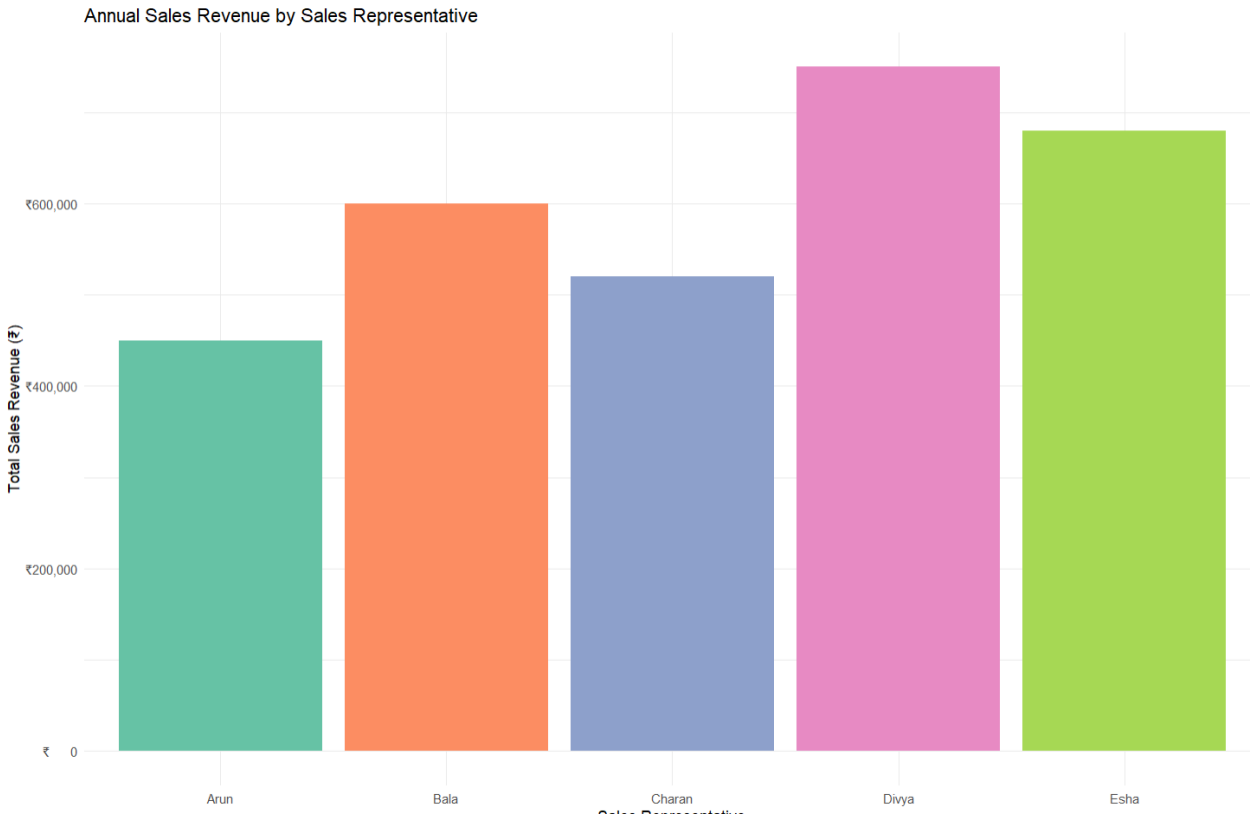
  Representative = c("Arun", "Bala", "Charan", "Divya", "Esha"),

  Revenue = c(450000, 600000, 520000, 750000, 680000)

)

ggplot(sales, aes(x = Representative, y = Revenue,
```

```
            fill = Representative)) +
geom_col() +
labs(
  title = "Annual Sales Revenue by Sales Representative",
  x = "Sales Representative",
  y = "Total Sales Revenue (₹)"
) +
scale_y_continuous(labels = function(x) {
  paste0("₹", format(x, big.mark = ",", scientific = FALSE))
}) +
scale_fill_brewer(palette = "Set2") +
theme_minimal() +
theme(legend.position = "none")
```



Interpretation:

- Divya generates the highest revenue of ₹7,50,000.
- Arun generates the lowest revenue of ₹4,50,000.
- The bar heights make it easy to compare the sales performance of all five representatives.

Conclusion: A bar chart is suitable for comparing revenue values across different categories.

Q3. Distributions – Daily Employee Commute Distances

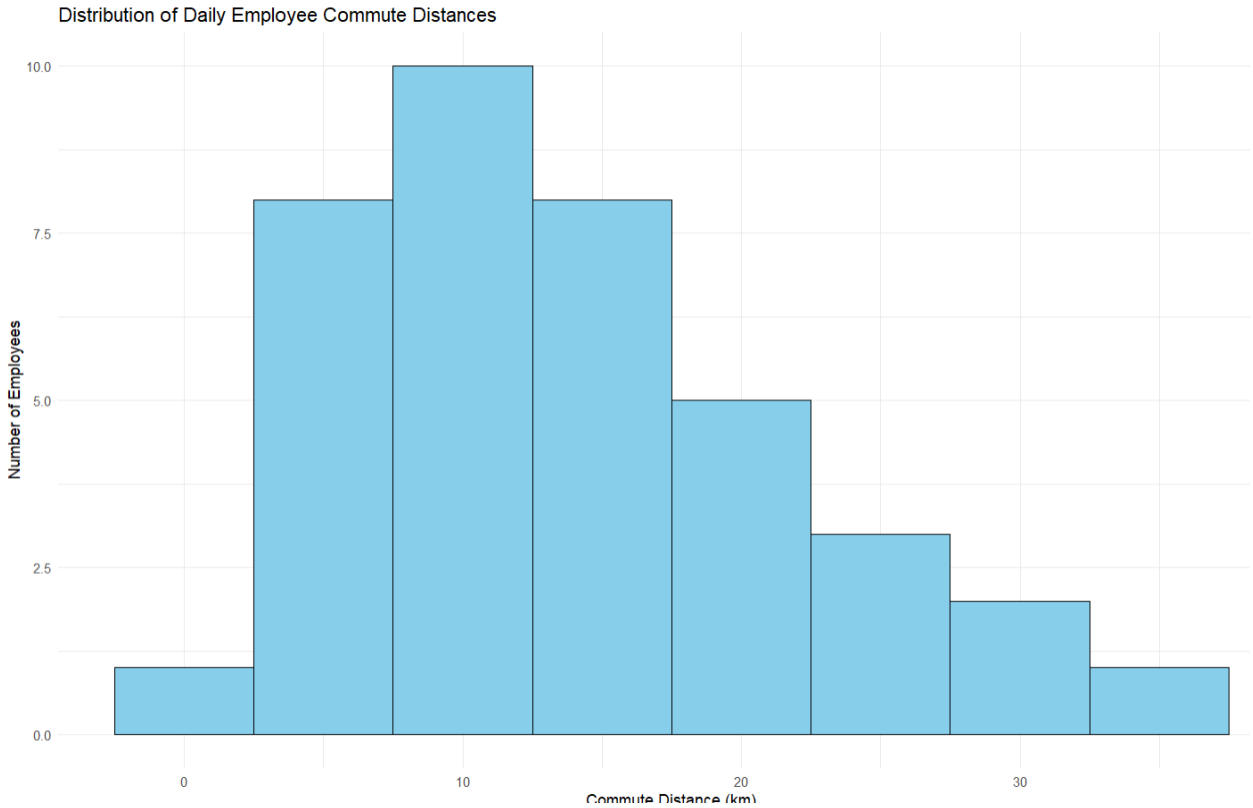
Aim: Represent the distribution of daily commute distances for 38 IT company employees using a histogram.

```
library(ggplot2)

distance <- data.frame(
  Distance = c(
    2, 3, 4, 5, 5, 6, 6, 7, 7, 8,
    8, 9, 9, 10, 10, 11, 11, 12, 12,
    13, 13, 14, 14, 15, 15, 16, 17,
    18, 18, 20, 21, 22, 23, 25, 26,
    28, 30, 35
  )
)

ggplot(distance, aes(x = Distance)) +
  geom_histogram(
    binwidth = 5,
    fill = "skyblue",
    color = "black"
  ) +
  labs(
    title = "Distribution of Daily Employee Commute Distances",
```

```
  x = "Commute Distance (km)",
  y = "Number of Employees"
) +
theme_minimal()
```



Interpretation:

- The histogram groups commute distances into intervals of 5 km.
- The tallest bars indicate the distance intervals containing the most employees.
- The chart helps identify the common commute range and whether the distribution is skewed.

Conclusion: A histogram is appropriate for showing the frequency distribution of continuous numerical data such as commute distance.

Q4. Proportions – Workforce Across Four Age Groups

Aim: Visualize the percentage distribution of a company's workforce across four age groups using a pie chart.

```
library(ggplot2)

workforce <- data.frame(

  AgeGroup = c("18–25", "26–35", "36–45", "Above 45"),

  Percentage = c(25, 40, 22, 13)

)

ggplot(workforce, aes(x = "", y = Percentage,

                      fill = AgeGroup)) +

  geom_col(width = 1, color = "white") +

  coord_polar(theta = "y") +

  geom_text(

    aes(label = paste0(Percentage, "%")),

    position = position_stack(vjust = 0.5)

  ) +

  labs(

    title = "Percentage Distribution of Workforce by Age Group",

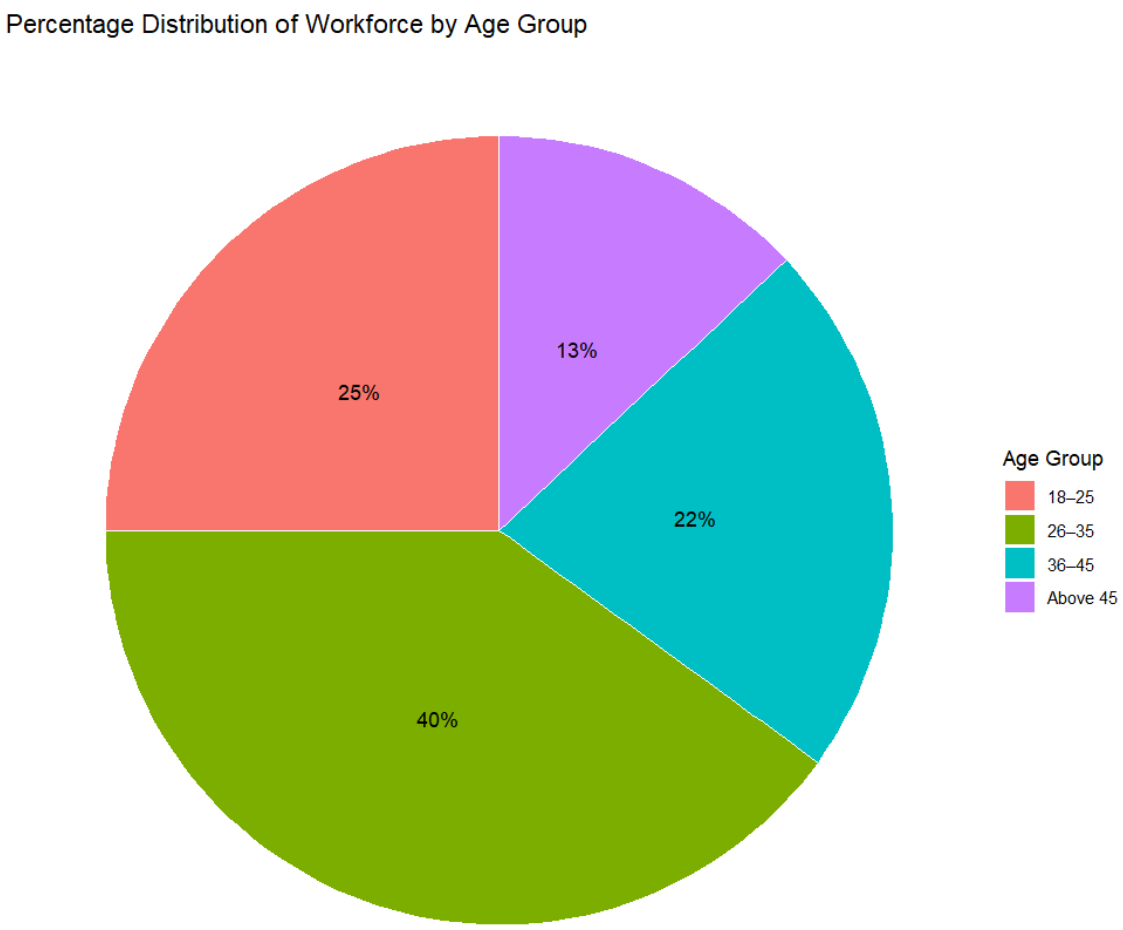
    fill = "Age Group"

  ) +

  theme_void()
```

Interpretation:

- Employees aged 26–35 form the largest group at 40%.
- Employees above 45 form the smallest group at 13%.
- Each slice represents that age group's share of the total workforce.



Conclusion: A pie chart is suitable for showing how a whole workforce is divided into percentage-based categories. The percentages total 100%.

Q5. X–Y Relationships – Marketing Spend and New Customers

Aim: Show the relationship between monthly marketing expenditure and the number of new customers acquired over 20 months using a scatter plot.

```
library(ggplot2)

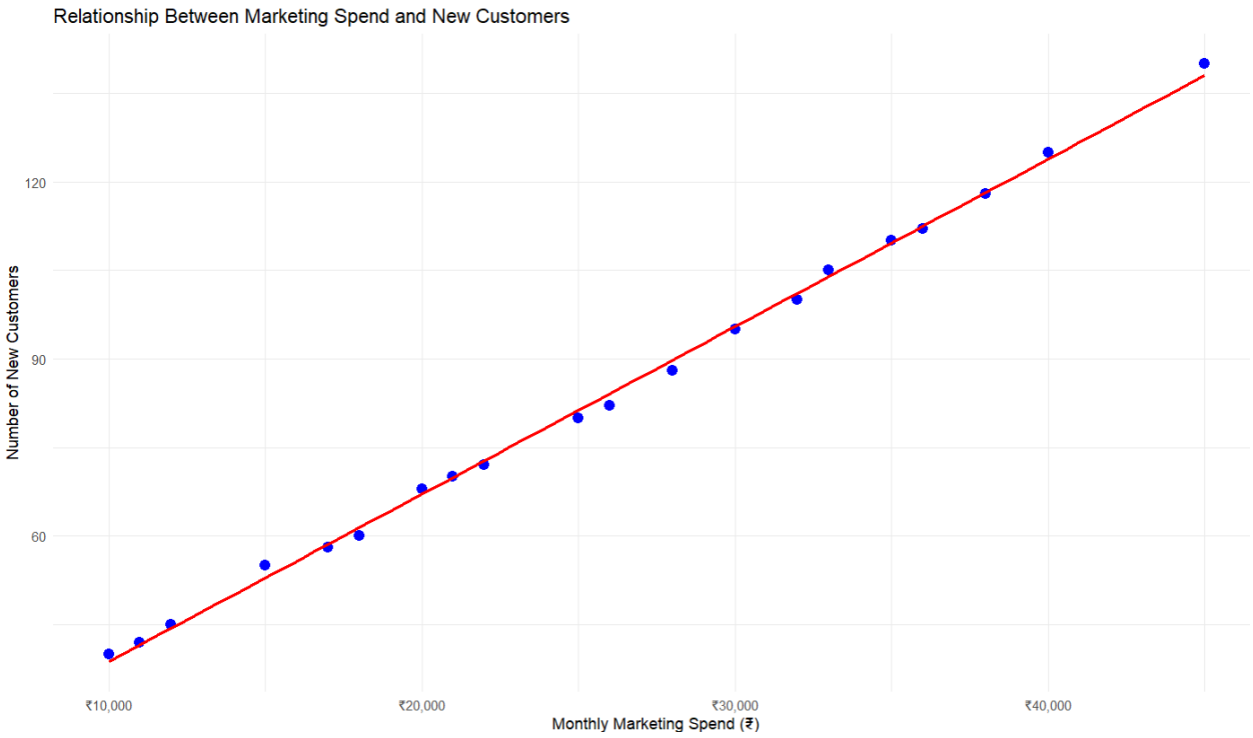
marketing <- data.frame(
  Month = 1:20,
  MarketingSpend = c(
    10000, 12000, 15000, 11000, 18000,
    20000, 17000, 22000, 25000, 21000,
```

```
    28000, 30000, 26000, 32000, 35000,
    33000, 38000, 40000, 36000, 45000
  ),
  NewCustomers = c(
    40, 45, 55, 42, 60,
    68, 58, 72, 80, 70,
    88, 95, 82, 100, 110,
    105, 118, 125, 112, 140
  )
)
ggplot(marketing, aes(x = MarketingSpend,
                      y = NewCustomers)) +
  geom_point(color = "blue", size = 3) +
  geom_smooth(method = "lm", se = FALSE,
              color = "red") +
  labs(
    title = "Relationship Between Marketing Spend and New Customers",
    x = "Monthly Marketing Spend (₹)",
    y = "Number of New Customers"
  ) +
  scale_x_continuous(labels = function(x) {
    paste0("₹", format(x, big.mark = ",", scientific = FALSE))
  }) +
  theme_minimal()
```

Interpretation:

- Each point represents one month's marketing expenditure and the number of new customers acquired.

- The upward-sloping red trend line indicates a positive relationship in the sample data.
- Higher marketing expenditure is generally associated with more new customers.
- The chart shows association, not proof that increased spending alone causes customer growth.



Conclusion: A scatter plot is appropriate for examining the relationship between two numerical variables. The trend line helps identify the overall direction of the relationship.